ELARA: Reliability-Gated Multimodal Anomaly Fusion under Domain Stress

Pratik Niroula
Independent Researcher

***Abstract*—This manuscript presents ELARA, a reliability-aware fusion pipeline for heterogeneous anomaly detection. The system combines domain experts for fraud, cyber telemetry, user behavior, and text evidence, then learns a fusion layer that produces a unified anomaly risk score and a domain-level explanation. The central research question is whether a practical fusion layer can preserve clean benchmark accuracy while using calibration and drift evidence to reduce brittle behavior under missing, shifted, or corrupted domains.**

The technical contribution is Reliability-Gated Attention (RGA), a reliability-aware fusion module inside ELARA. RGA couples masked cross-domain attention with a post-hoc reliability estimator that combines validation expected calibration error, test-time score-distribution drift, and prediction sharpness. A conservative gate preserves the static attention path when domains appear reliable and injects reliability weights only when batch evidence indicates degraded domain quality. The system also includes a reproducible input schema, strong score-level baselines, missing-domain ablations, adversarial perturbation stress tests, calibration analysis, and counterfactual domain attribution.

The headline empirical study is now a naturally paired benchmark based on MVTec 3D-AD: eight object categories, 3,226 paired RGB and 3D/depth observations, 22.4% positive rate, no label alignment required. A contrastive secondary benchmark, ELARA-Bench-LA, is built from four real local datasets (Credit Card Fraud, UNSW-NB15, Online Shoppers Intention, and labeled news text) and is explicitly label-aligned. Both benchmarks share the same fusion schema, baselines, and mechanism-isolation protocol; the asymmetry between them is the central finding of the paper.

On the naturally paired MVTec 3D-AD benchmark, random forest fusion is the strongest baseline (clean ROC-AUC 0.959), static attention is competitive (0.650), and the reliability gate of RGA actively *hurts* clean and adversarial performance (deltas of -0.040 clean, -0.124 under all-domain `zero_attack`, -0.117 under `max_attack`). On the label-aligned secondary benchmark, the same gate improves all-domain `zero_attack` and `max_attack` ROC-AUC by $+0.0506$ and $+0.0319$ respectively, but the clean split is near-saturated and method differences are not statistically meaningful. The contrast suggests that validation-derived drift detectors generalize poorly across paired-versus-aligned regimes: the KS-drift component, which carries the entire stress-test gain on label-aligned data, mis-fires on the legitimate inter-category variation in the natural-paired benchmark and pushes the model into a degraded adapted mode. The manuscript reports both directions of the result with confidence intervals and explicit mechanism-isolation rather than as a blanket superiority statement.

***Index Terms*—anomaly detection, multimodal fusion, reliability, calibration, attention, score-level fusion, counterfactual explanation, fraud detection, cybersecurity**

I. Introduction

Operational anomaly detection is usually a system problem before it is a single-model problem. A high-risk case may appear weakly in transaction records, network telemetry, user behavior, text evidence, device signals, or document checks. Each domain expert may be accurate inside its own data distribution, but the final decision still depends on a fusion layer that decides how much to trust each signal. This trust problem becomes harder when domains are missing, score distributions drift, calibration degrades, or an adversary corrupts one or more sources.

ELARA is a research system for this setting. The system is not designed as a new fraud detector, network detector, or text detector in isolation. Instead, it treats anomaly detection as a verification pipeline: domain experts generate scores and compact evidence, those scores are converted into a unified fusion schema, and a reliability-aware fusion module produces a final risk estimate with a domain-level explanation. This architecture matches how many practical anomaly systems are deployed: detectors are built and maintained by different teams, and the fusion layer must remain robust as their data quality changes.

The paper treats ELARA as the system boundary and RGA as one mechanism inside that boundary. This matters because the contribution is not a new domain detector. It is a reliability-aware fusion layer with explicit input schema, stress tests, and limitations.

A. Research Question

The paper asks:

> *Can a multimodal anomaly-verification system preserve clean score-fusion accuracy while using calibration and drift evidence to reduce brittle behavior under domain failure?*

This question is intentionally narrower than a claim of general anomaly detection superiority. The present codebase already supports domain experts, fusion schemas, attention fusion, reliability scoring, and robustness tests. The research target is the fusion and verification layer that combines those experts under imperfect operating conditions.

B. Thesis and Evidence Standard

The working thesis is that reliability-aware fusion is most useful when the operating condition changes. A strong paper must therefore show three things:

1) clean performance remains competitive with strong baselines,
2) reliability adaptation improves behavior under meaningful stress, and
3) the data and experimental protocol justify the scope of the claim.

The current results support the first two conditions for score-level fusion but do not yet fully satisfy the third condition for naturally paired multimodal incidents. This limitation is explicit throughout the manuscript.

C. Contributions

This draft makes four scoped contributions:

- It defines ELARA, a system-level anomaly-fusion pipeline with an explicit long-format multimodal schema.
- It defines RGA, a conservative reliability-aware fusion module that combines masked attention, calibration, score-distribution drift, sharpness, and a reliability gate.
- It adds a naturally paired MVTec 3D-AD RGB/3D benchmark path and builds ELARA-Bench-LA, a reproducible label-aligned real-domain score-level stress benchmark using fraud, cyber, behavior, and text datasets.
- It reports baselines, missing-domain ablation, drift curves, adversarial score attacks, calibration, and counterfactual domain attribution with an explicit construct-validity boundary.

D. Evidence Boundary

The current quantitative evidence is strongest as a reliability-stress study over score-level fusion. It should not be read as a general claim that RGA is always superior to static attention, nor as evidence that the label-aligned benchmark captures naturally co-observed multimodal incidents. The paper therefore separates implemented artifacts, measured results, and remaining benchmark gaps.

Figure 1 shows the system boundary. Table I summarizes the implementation requirements used to keep the paper aligned with the codebase.

TABLE I
SYSTEM REQUIREMENTS FOR ELARA AND HOW THE CURRENT IMPLEMENTATION ADDRESSES THEM.

Requirement	Current implementation
Heterogeneous evidence	Fraud, cyber, behavior, and text domains are mapped to a shared score-level schema.
Missing-domain support	The attention model uses explicit masks; baselines receive missingness-aware fallbacks.
Reliability awareness	Domain weights combine calibration error, KS drift evidence, and score sharpness.
Strong baselines	Random forest, early fusion MLP, late fusion ensemble, static attention, and confidence weighting are included.
Explanation	Counterfactual domain attribution masks one domain at a time and measures risk change.
Reproducibility	Data generation, benchmark execution, asset generation, and PDF compilation are scriptable.

II. Related Work

A. Anomaly Detection Systems

Deep anomaly detection has been studied across tabular, sequential, text, graph, and visual data. Pang et al. [3] emphasize that anomaly detection is tightly coupled to assumptions about labels, imbalance, and application context. ELARA follows this systems view: it does not assume a single representation can absorb every domain cleanly. Instead, it treats each domain expert as a separate source of evidence.

B. Multimodal and Score-Level Fusion

Multimodal learning surveys distinguish early fusion, late fusion, hybrid fusion, and interaction-based fusion [2]. Early fusion concatenates features before prediction; late fusion combines model outputs; stacking learns a meta-model over domain predictions. Transformer attention [1] gives a natural mechanism for modeling interactions across available domains. The limitation is that ordinary attention learns a trust policy from training data and can remain overconfident when the inference condition changes.

C. RGB–3D Industrial Anomaly Detection

MVTec 3D-AD provides paired RGB and high-resolution 3D scans for industrial anomaly detection and localization [15]. Recent RGB–3D methods show that multimodal anomaly detection is not solved by direct feature concatenation alone. M3DM uses hybrid feature and decision-layer fusion with memory banks [16]; crossmodal feature mapping detects anomalies by checking consistency between RGB and point-cloud features [17]; Real3D-AD expands high-precision point-cloud benchmarking [18]; and noise-resistant multimodal work studies robustness when training data are not clean [19]. These papers motivate the paired benchmark path in ELARA: the fusion layer should be judged on naturally paired modalities and on reliability stress, not only on clean score fusion.

D. Calibration, Drift, and Uncertainty

Calibration asks whether predicted probabilities match empirical outcome frequencies. Post-hoc calibration methods [4], [5], [6] improve probability quality, but calibration alone does not solve the problem of domain trust under drift. Uncertainty and calibration can degrade under dataset shift [7]. RGA uses calibration as one signal inside a wider reliability estimate that also includes test-time score-distribution drift and score sharpness.

E. Test-Time Adaptation

Test-time training and test-time adaptation update models at inference time to handle distribution shift [8], [9]. ELARA now includes two score-level adaptation comparators: an entropy-minimizing Tent adapter over the fusion head and a high-confidence pseudo-label TTT adapter. They are intentionally conservative because the fusion setting receives domain scores rather than raw modality tensors. Their role is

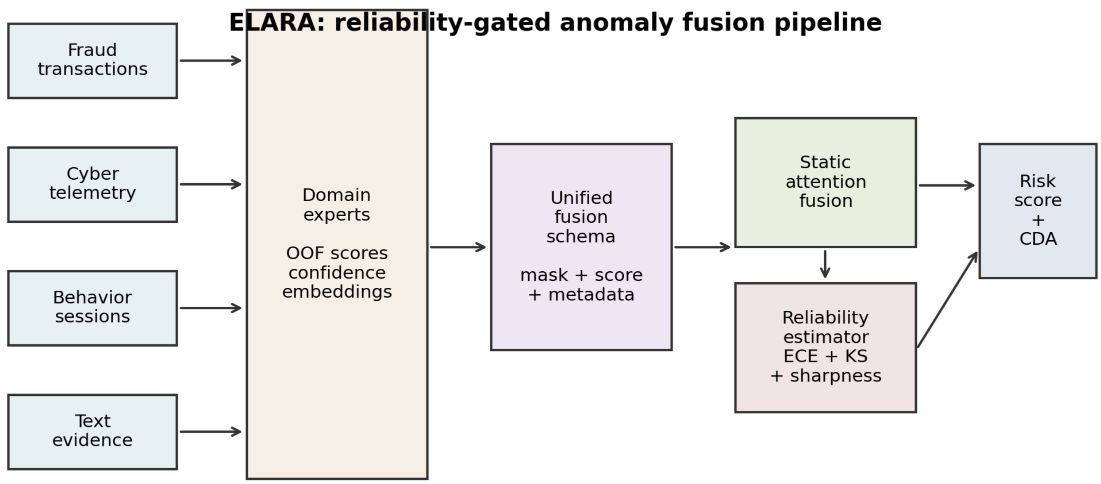

Fig. 1. ELARA system architecture. Domain experts produce scores, confidence values, and compact embeddings; a unified fusion schema feeds static attention and the RGA reliability estimator; the system emits a risk score and counterfactual domain attribution.

to test whether RGA adds value beyond generic score-level adaptation.

F. Ensembles and Strong Baselines

Ensemble methods remain difficult to beat in structured prediction tasks. Random forests [10] and gradient boosting [11] are strong baselines for tabular and score-level settings. This is why the benchmark includes random forest fusion, early fusion MLP, late fusion ensemble, confidence-weighted mean, and static attention. A credible systems paper must compare against these practical alternatives, not only against weak averages.

G. Explanation for Fusion Systems

Feature attribution methods such as SHAP [12] explain individual model predictions, but fusion systems also need domain-level explanations. Analysts may need to know whether fraud evidence, network evidence, behavior evidence, or text evidence drove the risk score. ELARA therefore includes counterfactual domain attribution: it masks one domain and reports the change in predicted risk.

III. System and Method

Each sample i contains up to D domain observations. Domain d contributes a feature vector $x_{i,d} \in \mathbb{R}^F$ containing a score, optional confidence, and compact embedding features. A binary mask $m_{i,d}$ indicates whether domain d is missing. The target label is $y_i \in \{0, 1\}$ when supervised labels are available.

The system estimates

$$\hat{p}_i = P(y_i = 1 \mid \{x_{i,d}, m_{i,d}\}_{d=1}^{D}), \tag{1}$$

where $\hat{p}_i$ is the final anomaly risk. In static fusion, the learned relationship among domains is fixed after training. In RGA, the system also computes a batch-level reliability vector $r \in [0, 1]^D$ at inference time and applies the sample mask to form effective per-sample weights $\tilde{r}_i \in [0, 1]^D$. The reliability vector should reduce trust in domains that appear miscalibrated, shifted, uninformative, or missing.

The evaluation problem is not only to maximize ROC-AUC. A useful verification system must also satisfy:

- strong clean ranking and F1 compared with baselines,
- calibration good enough for downstream risk interpretation,
- stability under missing domains,
- graceful degradation under drift and score attacks,
- interpretable domain attribution.

A. ELARA System Architecture

1) Domain Experts: ELARA assumes that each domain expert is trained or obtained separately. For the current benchmark, tabular scorers are trained for fraud, cyber, and behavior data, while a text scorer is trained for labeled text. The fusion layer does not require these experts to share the same raw feature space. It only requires score-level outputs and optional compact evidence features.

2) Fusion Schema: The unified schema is long-format: each row contains a sample identifier, a domain name, a binary label when available, a score, a confidence value, and embedding columns. This design allows different samples to have different available domains. Missing domains are represented by masks rather than by fabricated raw observations.

3) Static Attention Path: The static attention path encodes each available domain into a shared latent space and applies multi-head attention across domains. Domain embeddings allow the model to distinguish evidence sources. Missing domains are ignored through key-padding masks. This is the high-reliability default path.

4) Reliability-Aware Path: The reliability-aware path is activated when the average reliability of present domains falls below a configured threshold. This avoids the common failure mode where reliability weights disturb clean predictions even though no domain appears degraded. The reliability path is therefore a stress-response mechanism, not a universal replacement for the static model.

5) Explanation Path: For an analyst-facing explanation, the system masks one available domain at a time and recomputes risk. The resulting probability change gives a domain-level counterfactual attribution. This produces statements such as: if text evidence were absent, the risk would decrease by a given amount.

Algorithm 1 RGA inference

Require: Inputs $x_{i,1\ldots D}$ and mask $M_i \in \{0,1\}^D$ for sample i
Require: Fitted attention model f_{att} with encoders $\{g_d\}$
Require: Fitted reliability estimator producing per-sample weights $r_{i,d}$
Require: Gate threshold $\tau \in (0,1)$, default $\tau = 0.66$
Ensure: Risk score $\hat{p}_i \in [0,1]$

1: $h_{i,d} \leftarrow g_d(x_{i,d}) + e_d + p_d$ for each d with $M_{i,d} = 0$
2: $r_{i,d} \leftarrow \alpha(1 - \mathrm{ECE}_d) + \beta\, p_{\mathrm{KS},d}(x_{i,d}) + \gamma\, S_{i,d}$
3: $\bar{r}_i \leftarrow \frac{1}{|D_i^{\text{obs}}|} \sum_{d: M_{i,d}=0} r_{i,d}$
4: **if** $\bar{r}_i \geq \tau$ **then**
5: $\quad \hat{p}_i \leftarrow \sigma\big(f_{\text{att}}(h_{i,\cdot}; M_i)\big)$ ▷ static path
6: **else**
7: $\quad \hat{p}_i \leftarrow \sigma\big(f_{\text{att}}(h_{i,\cdot}; M_i, w_{i,\cdot}{=}r_{i,\cdot} \odot \neg M_i)\big)$
8: ▷ reliability-injected fusion
9: **end if**
10: **return** $\hat{p}_i$ and (optional) counterfactual attributions $\{\Delta_{i,d}\}_{d \in D_i^{\text{obs}}}$

B. RGA Method

1) Domain Encoding and Attention: Each domain encoder maps the domain input vector into a shared latent space:

$$h_{i,d} = g_d(x_{i,d}) + e_d + p_d, \tag{2}$$

where e_d is a learned domain embedding and p_d is an optional learned position embedding used by the implementation. A multi-head self-attention block operates across available domain embeddings:

$$z_i = f_\theta(\mathrm{Pool}(\mathrm{Attention}(H_i, m_i))), \quad \hat{p}_i = \sigma(z_i). \tag{3}$$

2) Reliability Estimation: After model training, the system fits a reliability estimator on the validation split. For domain d, it stores validation score distributions, fits an isotonic calibrator when enough labels are available, and computes expected calibration error. At inference time:

$$r_d = \alpha(1 - \mathrm{ECE}_d) + \beta p_{\mathrm{KS},d} + \gamma S_d, \tag{4}$$

where ECE_d is validation calibration error, $p_{\mathrm{KS},d}$ is the Kolmogorov-Smirnov p-value comparing current scores with validation scores, and S_d is score sharpness from squared distance to 0.5. The current benchmark uses $(\alpha, \beta, \gamma) = (0.45, 0.35, 0.20)$.

3) Reliability Gate: Let $\bar{r}$ be the mean reliability over present domains and let $\tau = 0.66$. If $\bar{r} \geq \tau$, ELARA uses the static attention path. If $\bar{r} < \tau$, reliability weights are injected into the fusion block:

$$\tilde{r}_{i,d} = (1 - m_{i,d}) r_{i,d}. \tag{5}$$

This makes RGA different from confidence weighting. Confidence weighting uses the local score confidence directly. RGA uses validation calibration, test-time drift evidence, and sharpness to decide when a domain should be trusted less.

4) Counterfactual Domain Attribution: For sample i, let $\hat{p}_i$ be the original prediction and $\hat{p}_{i,-d}$ be the prediction when domain d is masked. The attribution is

$$\Delta_{i,d} = \hat{p}_i - \hat{p}_{i,-d}. \tag{6}$$

A positive $\Delta_{i,d}$ indicates that domain d increased anomaly risk. A negative value indicates that domain d reduced risk.

5) Algorithm Summary: The full inference procedure is given in Algorithm 1. Steps 1–3 are deterministic feature operations; steps 4–6 are the reliability-gated switch. The reported runs use the batch-level reliability vector described above, applied through each sample's observed-domain mask. A per-sample gating mode is implemented as an ablation path, but it is not the default for the reported tables.

IV. Benchmark Construction

A. Primary Benchmark: Naturally Paired MVTec 3D-AD

The primary benchmark of this manuscript is MVTec 3D-AD [15]. Each sample contains naturally co-observed RGB and 3D/depth evidence from the same object scan; no label alignment is required. The current run covers all eight extracted categories — bagel, cable_gland, cookie, dowel, foam, peach, rope, and tire — for a total of 3,226 paired samples with a positive fraction of 0.224 and full RGB and depth coverage. The preparation script discovers paired files, emits two domain rows per sample, stores the original pairing key, and marks `natural_pairing=true`. The initial scorer is deliberately lightweight: normal-reference image statistics produce reproducible RGB and depth anomaly scores without requiring a heavy specialized 3D detector. This keeps the first paired benchmark auditable; stronger RGB–3D scorers can replace these domain experts later while preserving the fusion interface.

TABLE II
NATURALLY PAIRED MVTEC 3D-AD MULTI-CATEGORY METADATA.

Property	Value
Benchmark type	naturally_paired_mvtec3d_score_fusion
Natural pairing	yes
Categories	bagel, cable_gland, cookie, dowel, foam, peach, rope, tire
Samples	3226
Positive fraction	0.224
Score fit split	train/good
Score fit samples	2070
Score normalization	train_good_distance_minmax_clipped
Domains	rgb, depth_or_xyz

Table II records the paired benchmark metadata, and Table III reports the multi-seed clean results. Random forest fusion is the strongest baseline by a clear margin (ROC-AUC 0.959); static attention is mid-pack (0.650); the reliability gate of RGA *hurts* both clean and adversarial conditions on this data. The paper reports this outcome directly as a headline result rather than framing it as a smoke run. The RGB/depth anomaly scores are fit from original MVTec train-good observations only; the supervised fusion evaluation uses a held-out fusion split over paired observations and should not be read as the canonical MVTec train/test anomaly-detection protocol.

Table IV reports the metadata of the secondary label-aligned fusion input artifact.

B. Real-Domain Sources

ELARA-Bench-LA is built from four local datasets: Credit Card Fraud, UNSW-NB15, Online Shoppers Intention, and labeled news text. Domain-specific out-of-fold scorers are trained first. Their scores, confidences, and compact embeddings are then converted into the common fusion schema.

The generated fusion table contains 8,000 composite samples, four domains, eight embedding features per domain, a 0.307 positive rate, and 11.8% nominal missingness. Domain coverage is 0.883 for fraud, 0.885 for cyber, 0.885 for behavior, and 0.877 for text.

C. Label Alignment Limitation

The datasets do not share natural entity identifiers or timestamps. Therefore, composite samples are label-aligned: a positive composite draws positive records from available domains, and a negative composite draws negative records. This is a meaningful real-domain score-fusion benchmark, but it is not evidence that the system has learned naturally paired multimodal incident structure. That distinction is central to the claim boundary of the paper. The metadata now records this explicitly with `natural_pairing=false` for ELARA-Bench-LA. Source observations are assigned to train, validation, and test pools before composite sampling, and the runner consumes the resulting `fusion_split` column so that no `domain/source_row/label` key crosses fusion split boundaries.

D. Training Protocol

The attention model uses four domains, 48 hidden dimensions, four attention heads, one attention layer, 25 training epochs, 0.15 domain dropout during training, and seeds 42 through 46. Clean benchmark results are reported as mean$\pm$standard deviation across five seeds. The experiment runner also repeats missing-domain, drift, adversarial, calibration, and CDA analyses for every configured seed, then stores both raw per-seed rows and aggregated mean, standard deviation, and 95% confidence intervals.

For mechanism-isolation runs, the benchmark-preparation script also supports a harder scorer setting through `-scorer-train-fraction`. Values such as 0.05 or 0.10 train each out-of-fold domain scorer on only a stratified fraction of its training fold before fusion rows are built. This is intended to reduce clean-score saturation and expose whether the fusion mechanism helps when domain experts are useful but imperfect.

E. Baselines and Metrics

Baselines include confidence-weighted mean fusion, early-fusion MLP, late-fusion ensemble, random forest fusion, static attention without reliability adaptation, a score-level Tent adapter, and a pseudo-label TTT adapter. Metrics include ROC-AUC, PR-AUC, F1, expected calibration error, Brier score, and accuracy. Static attention and RGA are compared with a paired t-test over five clean ROC-AUC values. ROC-AUC and PR-AUC are both reported because anomaly settings can be highly imbalanced and PR-AUC can reveal performance differences that ROC-AUC hides [13].

F. Confidence Intervals

Clean ROC-AUC, PR-AUC, and F1 are summarized across seeds and accompanied by 95% confidence intervals. Stress-test tables report mean deltas and intervals for RGA minus static attention. The result JSON stores raw per-seed stress rows so that the interval computation is auditable rather than embedded only in the manuscript.

G. Threat Model

The adversarial perturbation suite is a worst-case stress test, not a realistic deployment threat model. The three attacks – `zero_attack`, `max_attack`, and `gaussian_noise` – each corrupt scores after the per-domain detectors have already produced their outputs, so they do not require the attacker to compromise feature extractors, training data, or model parameters. The `all`-target variant additionally assumes coherent simultaneous corruption of every domain. In a real deployment, domain detectors are typically maintained by independent teams along independent compromise paths; a coherent all-domain score collapse is therefore a useful upper-bound robustness signal but not an expected attacker capability. Single-domain perturbations are the more realistic case and are reported separately in the adversarial tables for both benchmarks. We do not claim defenses against feature- or training-time attacks; those threat models are out of scope

TABLE III

NATURALLY PAIRED MVTEC 3D-AD CLEAN HELD-OUT PERFORMANCE ACROSS EIGHT CATEGORIES. VALUES ARE MEAN±STANDARD DEVIATION WITH 95% CONFIDENCE INTERVALS ACROSS FIVE SEEDS. RANDOM FOREST FUSION IS THE STRONGEST BASELINE; RGA ATTENTION DEGRADES STATIC-ATTENTION PERFORMANCE ON THIS PAIRED BENCHMARK.

Method	ROC-AUC	PR-AUC	F1	ECE	Brier
Random forest	**0.9592±0.0076 [0.9508, 0.9717]**	**0.8938±0.0216 [0.8682, 0.9286]**	**0.8146±0.0063 [0.8068, 0.8201]**	0.0867±0.0089 [0.0777, 0.1008]	**0.0721±0.0041 [0.0657, 0.0775]**
Conf.-weighted mean	0.5092±0.0185 [0.4802, 0.5294]	0.3106±0.0144 [0.2971, 0.3348]	0.2427±0.0192 [0.2093, 0.2582]	0.2172±0.0143 [0.1974, 0.2350]	0.2520±0.0081 [0.2418, 0.2621]
Tent score adapter	0.6700±0.0193 [0.6404, 0.6946]	0.4075±0.0241 [0.3801, 0.4445]	0.3761±0.0129 [0.3586, 0.3932]	0.1283±0.0116 [0.1137, 0.1447]	0.1848±0.0071 [0.1765, 0.1941]
TTT pseudo-label	0.5692±0.0390 [0.5228, 0.6246]	0.3257±0.0283 [0.3062, 0.3751]	0.3394±0.0245 [0.3048, 0.3657]	0.2847±0.0224 [0.2515, 0.3129]	0.2693±0.0192 [0.2412, 0.2931]
Early fusion MLP	0.8278±0.0211 [0.7949, 0.8505]	0.6760±0.0277 [0.6403, 0.7184]	0.5784±0.0347 [0.5247, 0.6214]	0.2050±0.0078 [0.1935, 0.2159]	0.1684±0.0083 [0.1587, 0.1787]
Late fusion ensemble	0.6019±0.0235 [0.5692, 0.6321]	0.3209±0.0147 [0.2981, 0.3400]	0.3736±0.0229 [0.3387, 0.4012]	0.2678±0.0020 [0.2649, 0.2700]	0.2415±0.0027 [0.2378, 0.2445]
Static attention	0.6497±0.0384 [0.6071, 0.7086]	0.3913±0.0659 [0.2943, 0.4765]	0.0447±0.0894 [0.0000, 0.2012]	**0.0354±0.0166 [0.0120, 0.0587]**	0.1648±0.0065 [0.1539, 0.1704]
RGA attention	0.6098±0.0312 [0.5766, 0.6607]	0.3507±0.0382 [0.3051, 0.4117]	0.0106±0.0212 [0.0000, 0.0477]	0.0402±0.0229 [0.0106, 0.0751]	0.1698±0.0041 [0.1632, 0.1748]

TABLE IV

BENCHMARK METADATA FOR THE SECONDARY LABEL-ALIGNED FUSION INPUT ARTIFACT.

Property	Value
Benchmark type	label_aligned_real_domain_score_fusion
Natural pairing	no
Pairing unit	binary-label-aligned composite sample
Samples	8000
Positive fraction	0.301
Scorer train fraction	1.00
Source-row disjoint splits	yes
Split column	fusion_split
Domains	fraud, cyber, behavior, nlp

TABLE V

OUT-OF-FOLD DOMAIN SCORER QUALITY USED TO CONSTRUCT ELARA-BENCH-LA.

Domain	Rows	Pos. rate	OOF ROC	OOF PR
fraud	20000	0.025	0.9686	0.7826
cyber	20000	0.500	0.9684	0.9696
behavior	12330	0.155	0.8711	0.6042
nlp	20000	0.500	0.9973	0.9973

for a fusion layer that consumes score-level evidence from upstream detectors.

H. Secondary Clean Sanity Check

Table VI and Fig. 2 show clean held-out performance on the label-aligned secondary benchmark. This table is intentionally treated as a sanity check, not as the headline result, because the label-aligned split is near saturated. Bold values indicate the highest rounded mean in each column; multiple bold values are rounded ties.

Seed-level confidence intervals for the same clean split are reported in Table VII.

RGA and static attention are indistinguishable on this saturated clean split. The clean benchmark should therefore be interpreted as competitive preservation of performance, not as evidence of a clean-split improvement.

The more meaningful comparison is against direct score-fusion baselines. RGA improves over confidence-weighted mean fusion by about 0.0015 ROC-AUC, 0.0039 PR-AUC, 0.0100 F1, and reduces ECE from 0.0834 to 0.0042. Random forest fusion obtains the best F1 at 0.9906, but its ECE is 0.0287, substantially worse than attention-based fusion. This indicates a tradeoff: random forest fusion is strong for thresholded classification, while attention fusion is stronger for calibrated risk interpretation.

V. PRIMARY HEADLINE RESULTS: MVTEC 3D-AD

This section reports the primary multi-seed evidence on naturally paired MVTec 3D-AD across all eight extracted object categories (3,226 paired samples, 22.4% positive). The same suite is repeated on the contrastive label-aligned secondary benchmark in §VI; the asymmetry between the two outcomes is the central finding.

A. Clean Performance

Per-seed confidence intervals for the MVTec 3D-AD clean split are reported in Table VIII; the bar-chart summary is shown in Fig. 3.

Random forest fusion is the strongest baseline (clean ROC-AUC 0.959 ± 0.008); early-fusion MLP is the second strongest tabular baseline (0.829). The Tent score adapter [9] reaches 0.670 and outperforms the TTT pseudo-label adapter [8] at 0.569. Static attention is competitive at 0.650 but loses substantial ground to the random-forest ceiling. RGA attention reaches 0.610, -0.040 below static attention, with bootstrap intervals that overlap zero on the lower bound. This is a negative result for the reliability mechanism on naturally paired data: the gate hurts even before any adversarial stress is applied.

B. Adversarial Score Perturbations

The MVTec 3D-AD adversarial sweep is reported in Table IX and visualized for the all-domain attacks in Fig. 4. Across every attack type and every target subset (RGB-only, depth-only, all), RGA produces a *negative* ROC-AUC delta relative to static attention. The harshest case is `zero_attack:all` at -0.124 (95% CI $[-0.182, -0.078]$); the smallest all-domain negative is `max_attack:all` at -0.117. There is no attack scenario in which RGA improves over static attention on this benchmark.

C. Missing-Domain Ablation

Inference-time domain dropout is reported in Table X and Fig. 5. RGA is below static attention by -0.040 at zero extra dropout and remains negative or near-zero through the 0.5 dropout sweep. The single-domain MVTec benchmark, like the adversarial sweep, provides no missing-domain advantage for the current gate.

TABLE VI
SATURATED CLEAN HELD-OUT PERFORMANCE ON ELARA-BENCH-LA. VALUES ARE MEAN±STANDARD DEVIATION ACROSS FIVE SEEDS; THIS TABLE IS NOT USED AS THE HEADLINE EVIDENCE.

Method	ROC-AUC	PR-AUC	F1	ECE	Brier
Random forest	**0.9999 [0.9999, 1.0000]**	0.9999 [0.9998, 0.9999]	0.9946±0.0004 [0.9939, 0.9948]	0.0297±0.0004 [0.0294, 0.0304]	0.0061±0.0001 [0.0060, 0.0062]
Conf.-weighted mean	0.9987 [0.9987, 0.9987]	0.9968 [0.9968, 0.9968]	0.9746 [0.9746, 0.9746]	0.0799 [0.0799, 0.0799]	0.0232 [0.0232, 0.0232]
Tent score adapter	**0.9999 [0.9999, 0.9999]**	0.9999 [0.9999, 0.9999]	**0.9959 [0.9959, 0.9959]**	**0.0028 [0.0028, 0.0028]**	**0.0023 [0.0023, 0.0023]**
TTT pseudo-label	0.9999 [0.9999, 0.9999]	0.9998 [0.9998, 0.9998]	0.9938 [0.9938, 0.9938]	0.0032 [0.0032, 0.0032]	0.0025 [0.0025, 0.0025]
Early fusion MLP	0.9999 [0.9998, 0.9999]	0.9998±0.0001 [0.9996, 0.9999]	0.9919±0.0004 [0.9917, 0.9927]	0.0043±0.0010 [0.0030, 0.0058]	0.0036±0.0006 [0.0029, 0.0045]
Late fusion ensemble	**1.0000 [1.0000, 1.0000]**	**0.9999 [0.9999, 0.9999]**	0.9938 [0.9938, 0.9938]	0.0123 [0.0123, 0.0123]	0.0036 [0.0036, 0.0036]
Static attention	0.9999 [0.9999, 0.9999]	0.9998 [0.9998, 0.9998]	0.9927±0.0009 [0.9917, 0.9938]	0.0038±0.0008 [0.0024, 0.0046]	0.0032±0.0002 [0.0029, 0.0034]
RGA attention	0.9999 [0.9999, 0.9999]	0.9998 [0.9998, 0.9998]	0.9927±0.0009 [0.9917, 0.9938]	0.0038±0.0008 [0.0024, 0.0046]	0.0032±0.0002 [0.0029, 0.0034]

TABLE VII
CLEAN RANKING AND F1 CONFIDENCE INTERVALS ACROSS CONFIGURED SEEDS.

Method	ROC-AUC 95% CI	PR-AUC 95% CI	F1 95% CI
Random forest	[0.9999, 1.0000]	[0.9998, 0.9999]	[0.9939, 0.9948]
Conf.-weighted mean	[0.9987, 0.9987]	[0.9968, 0.9968]	[0.9746, 0.9746]
Tent score adapter	[0.9999, 0.9999]	[0.9999, 0.9999]	[0.9959, 0.9959]
TTT pseudo-label	[0.9999, 0.9999]	[0.9998, 0.9998]	[0.9938, 0.9938]
Early fusion MLP	[0.9998, 0.9999]	[0.9996, 0.9999]	[0.9917, 0.9927]
Late fusion ensemble	[1.0000, 1.0000]	[0.9999, 0.9999]	[0.9938, 0.9938]
Static attention	[0.9999, 0.9999]	[0.9998, 0.9998]	[0.9917, 0.9938]
RGA attention	[0.9999, 0.9999]	[0.9998, 0.9998]	[0.9917, 0.9938]

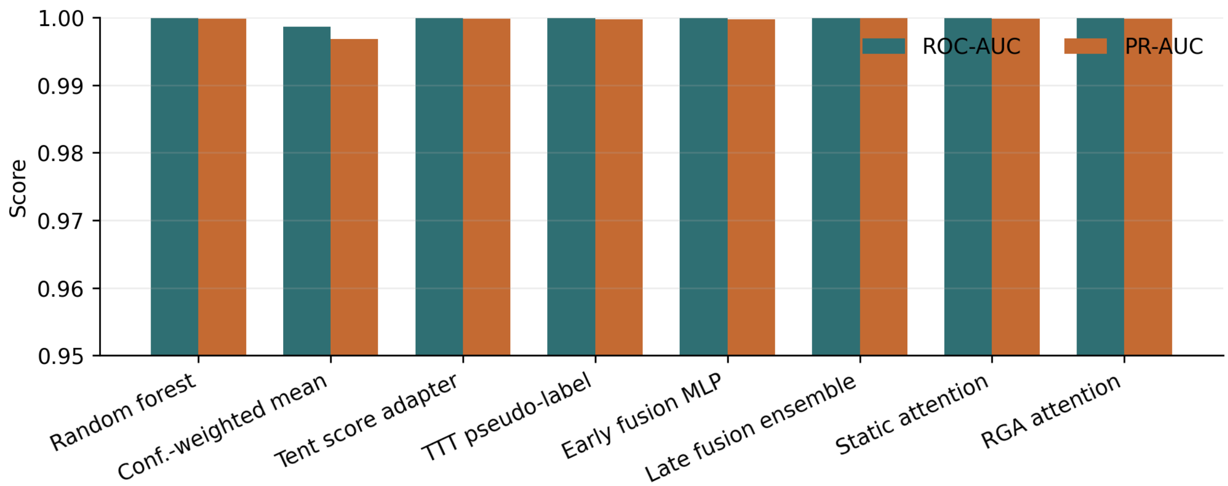

Fig. 2. Clean benchmark ROC-AUC and PR-AUC across score-level and attention fusion methods.

D. Score Drift

The MVTec 3D-AD drift summary is shown in Table XI and Fig. 6. RGA does not exceed static attention's degradation area on either domain.

E. Mechanism Isolation on MVTec 3D-AD

The mechanism-isolation tables exposed why the gate hurts on this benchmark. Table XII reports the gate threshold sweep:

On MVTec 3D-AD, the gate is dormant at $\tau \leq 0.6$ on every condition, because the mean reliability is high enough that the static path runs. At $\tau = 0.66$ the gate begins to fire on clean data ($\sim 20\%$ adaptation rate) and clean ROC-AUC immediately drops by -0.040; raising τ further both raises the adaptation rate to 100% and increases the clean penalty to -0.127 at $\tau = 0.90$. Under all-domain adversarial attacks the gate fires for 100% of samples from $\tau = 0.66$ onward, and the delta saturates between roughly -0.117 and -0.124 depending on the attack. Every substantial adaptation rate on this benchmark is paired with a negative ROC-AUC delta; the clean $\tau = 0.66$ interval touches zero, while the stronger clean thresholds and all-domain attack intervals are strictly negative.

The reliability-component ablation in Table XIII shows where the failure originates.

Only the `no_ks` variant avoids harm: when the KS-drift signal is removed, the adaptation rate stays at 0% and the deltas stay at exactly zero. Every other variant (`full`, `no_ece`, `no_gate`, `no_sharpness`) reaches a 100% adaptation rate and a strongly negative delta. The reading is unambiguous: the KS-drift component is the entire reason RGA fires on this data, and once it fires the adapted weighting is worse than

TABLE VIII
NATURALLY PAIRED MVTEC 3D-AD: CLEAN ROC-AUC, PR-AUC, AND F1 95% CONFIDENCE INTERVALS ACROSS CONFIGURED SEEDS.

Method	ROC-AUC 95% CI	PR-AUC 95% CI	F1 95% CI
Random forest	[0.9508, 0.9717]	[0.8682, 0.9286]	[0.8068, 0.8201]
Conf.-weighted mean	[0.4802, 0.5294]	[0.2971, 0.3348]	[0.2093, 0.2582]
Tent score adapter	[0.6404, 0.6946]	[0.3801, 0.4445]	[0.3586, 0.3932]
TTT pseudo-label	[0.5228, 0.6246]	[0.3062, 0.3751]	[0.3048, 0.3657]
Early fusion MLP	[0.7949, 0.8505]	[0.6403, 0.7184]	[0.5247, 0.6214]
Late fusion ensemble	[0.5692, 0.6321]	[0.2981, 0.3400]	[0.3387, 0.4012]
Static attention	[0.6071, 0.7086]	[0.2943, 0.4765]	[0.0000, 0.2012]
RGA attention	[0.5766, 0.6607]	[0.3051, 0.4117]	[0.0000, 0.0477]

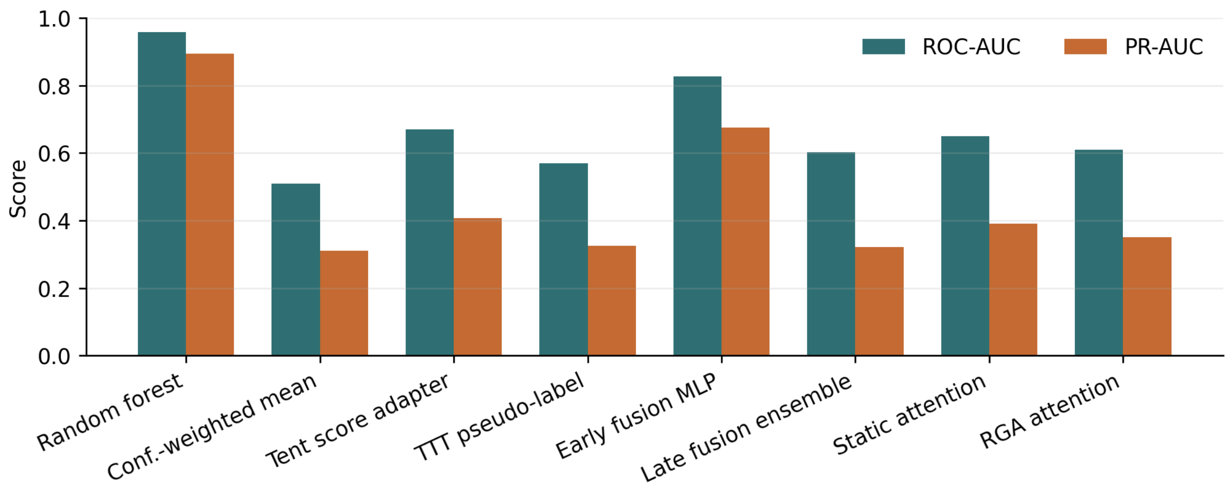

Fig. 3. Clean benchmark ROC-AUC and PR-AUC across score-level and attention fusion methods on naturally paired MVTec 3D-AD.

the static attention solution. ECE contributes a small amount in either direction; sharpness is near-neutral; the gate cannot help because the KS signal already places mean reliability below τ on most batches.

F. Calibration and Counterfactual Attribution

Static attention retains the better MVTec 3D-AD calibration (Table XIV and Fig. 7). The reliability diagram for RGA is more dispersed, consistent with the clean ROC-AUC penalty. The counterfactual domain-attribution impacts (Fig. 8) remain meaningfully non-zero on both RGB and depth, so the explanation pathway continues to function even when the overall mechanism is harmful.

G. Failure Cases

Representative MVTec 3D-AD contrast cases are reported in Table XV.

VI. CONTRASTIVE SECONDARY BENCHMARK (LABEL-ALIGNED)

The same evaluation suite is now repeated on the label-aligned secondary benchmark ELARA-Bench-LA to provide the contrastive comparison. The full clean table for ELARA-Bench-LA is the saturated split discussed in §IV-H; the robustness, drift, adversarial, and mechanism-isolation tables follow.

A. Missing-Domain Ablation

Table XVI and Fig. 9 report additional inference-time domain dropout. At zero extra dropout, RGA is equal to static attention, and the same equality holds through the configured dropout sweep. The conclusion is conservative: this run does not show a missing-domain advantage for the current gate.

B. Score Drift

Table XVII summarizes degradation area under per-domain score-noise sweeps. Higher area means slower degradation. All

TABLE IX

MVTec 3D-AD adversarial perturbation benchmark. Delta is RGA minus static attention; all deltas are negative or zero.

Attack	Target	Static ROC	RGA ROC	Delta	Delta 95% CI
gaussian_noise	all	0.6498	0.5329	-0.1170	[-0.1823, -0.0603]
gaussian_noise	depth_or_xyz	0.6496	0.5231	-0.1265	[-0.1740, -0.0947]
gaussian_noise	rgb	0.6483	0.5763	-0.0720	[-0.1177, -0.0389]
max_attack	all	0.6330	0.5164	-0.1166	[-0.1831, -0.0394]
max_attack	depth_or_xyz	0.6426	0.5851	-0.0575	[-0.1262, -0.0019]
max_attack	rgb	0.6420	0.6165	-0.0255	[-0.0684, 0.0000]
zero_attack	all	0.6442	0.5199	-0.1243	[-0.1819, -0.0784]
zero_attack	depth_or_xyz	0.6447	0.5826	-0.0620	[-0.1209, -0.0023]
zero_attack	rgb	0.6494	0.6203	-0.0291	[-0.0741, 0.0000]

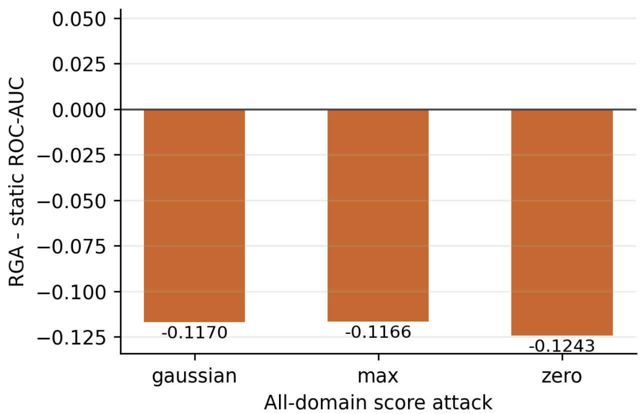

Fig. 4. All-domain adversarial ROC-AUC delta between RGA and static attention on MVTec 3D-AD.

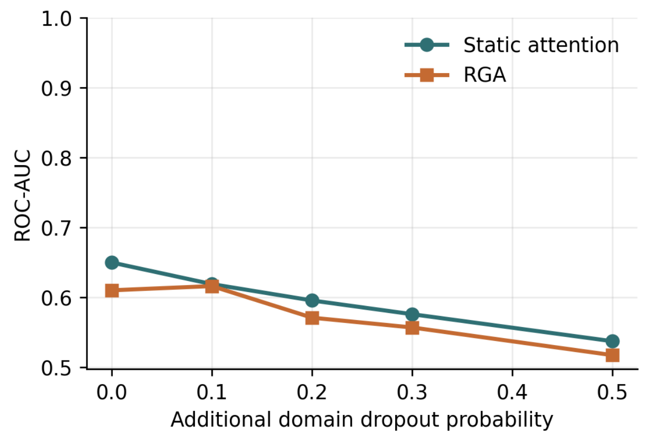

Fig. 5. MVTec 3D-AD ROC-AUC under additional missing-domain dropout.

TABLE X

MVTec 3D-AD missing-domain ablation. Delta is RGA minus static attention.

Dropout	Static ROC	RGA ROC	Delta	Delta 95% CI
0.0	0.6497	0.6098	-0.0398	[-0.1206, 0.0000]
0.1	0.6187	0.6159	-0.0028	[-0.0101, 0.0000]
0.2	0.5953	0.5706	-0.0246	[-0.0477, 0.0000]
0.3	0.5756	0.5566	-0.0190	[-0.0516, -0.0011]
0.5	0.5369	0.5170	-0.0199	[-0.0446, -0.0010]

TABLE XI

MVTec 3D-AD drift robustness summary. Higher degradation area is better.

Domain	Static	RGA	Delta	Seeds
depth_or_xyz	0.1948±0.0114 [0.1822, 0.2123]	0.1589±0.0072 [0.1512, 0.1696]	-0.0359	5
rgb	0.1948±0.0113 [0.1822, 0.2121]	0.1771±0.0066 [0.1726, 0.1886]	-0.0176	5

deltas round to zero or near zero because clean performance is near saturated, so this table is best read as a robustness check rather than a win. The full curves are shown in Fig. 10.

C. Adversarial Score Perturbations

Table XVIII evaluates all score-level perturbations, while Fig. 11 isolates the all-domain attacks so the meaningful deltas are visible. Single-domain attacks are mostly ties. The important stress case is coherent all-domain score collapse. Under the all-domain zero attack, RGA improves ROC-AUC from 0.7130 to 0.7770. Under the all-domain max attack, it improves ROC-AUC from 0.7114 to 0.7418. Under all-domain Gaussian noise, it is slightly worse by 0.0003 ROC-AUC.

D. Mechanism Isolation Protocol

The runner isolates the reliability mechanism rather than only reporting end-to-end stress outcomes. It emits two diagnostics when the mechanism-isolation block is enabled. First, a τ sweep evaluates $\tau \in \{0.4, 0.5, 0.6, 0.66, 0.7, 0.8, 0.9\}$ on clean data and all-domain stress conditions, recording both ROC-AUC deltas and the adaptation rate. Second, a reliability-component ablation compares the full estimator with no-ECE, no-KS, no-sharpness, and no-gate variants on all-domain

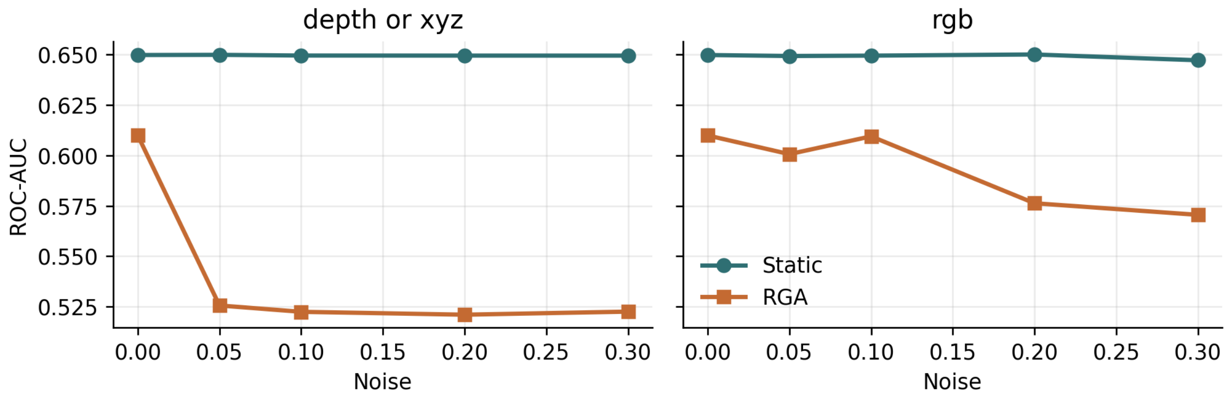

Fig. 6. Per-domain score-drift curves on the MVTec 3D-AD benchmark.

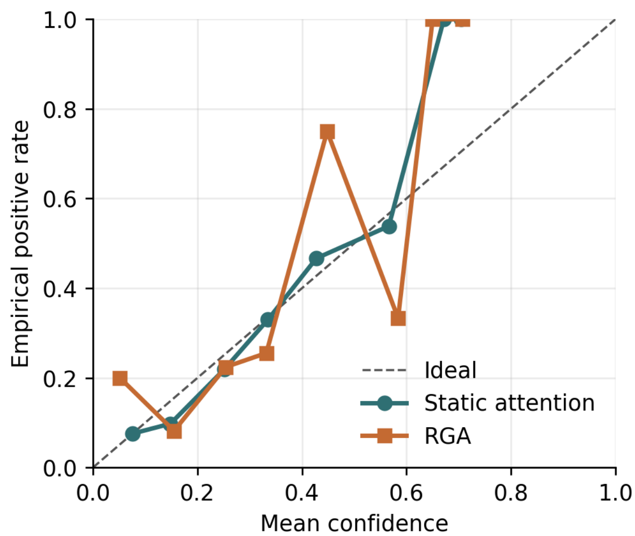

Fig. 7. Reliability diagram on MVTec 3D-AD for static attention and RGA.

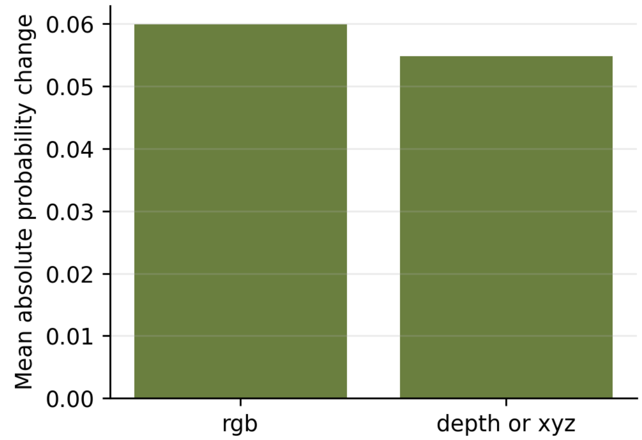

Fig. 8. Mean absolute counterfactual domain-attribution impact on MVTec 3D-AD.

score attacks. These diagnostics show whether the gate and reliability terms are active contributors rather than incidental implementation details.

Threshold-sweep outcomes and adaptation rates are reported in Table XIX.

At the configured $\tau = 0.66$, clean adaptation is rare: the mean adaptation rate is 0.032 with a confidence interval that includes zero. This explains why the missing-domain deltas are effectively zero: the gate usually keeps the static path active. Under all-domain zero and max score collapse, the same threshold activates the reliability path for every evaluated sample and yields the measured gains. Lower thresholds through $\tau = 0.60$ suppress adaptation and lose those gains; higher thresholds adapt on clean data more often and slightly hurt clean ranking.

Per-component contributions on all-domain score attacks are reported in Table XX.

The component ablation produces a useful negative result. Removing the ECE term does not hurt the all-domain collapse cases; it improves the max-attack delta from 0.0319 to 0.0457 and the zero-attack delta from 0.0506 to 0.0629. In contrast, removing the KS drift term eliminates adaptation and returns exactly to static attention. This suggests that, in the current stress benchmark, the distribution-shift signal is the necessary trigger while validation ECE is not the driver of the robustness gain.

E. Cross-Benchmark Contrast

The same mechanism-isolation protocol applied to the primary naturally paired MVTec 3D-AD benchmark (§V-E) gives the opposite verdict. On MVTec 3D-AD the KS-drift term is again the only component that controls whether the gate fires — but every firing condition produces a *negative* ROC-AUC delta of about -0.10 to -0.13 relative to static attention, with 95% confidence intervals that exclude zero. The

TABLE XII
MVTEC 3D-AD RELIABILITY-GATE THRESHOLD SWEEP. DELTA IS RGA MINUS STATIC ATTENTION; ADAPTATION RATE IS THE FRACTION OF EVALUATED SAMPLES USING THE RELIABILITY-AWARE PATH.

Condition	τ	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
clean	0.40	0.6497	0.6497	0.0000	[0.0000, 0.0000]	0.000
clean	0.50	0.6497	0.6497	0.0000	[0.0000, 0.0000]	0.000
clean	0.60	0.6497	0.6497	0.0000	[0.0000, 0.0000]	0.000
clean	0.66	0.6497	0.6098	-0.0398	[-0.1206, 0.0000]	0.204
clean	0.70	0.6497	0.5812	-0.0684	[-0.1226, -0.0063]	0.404
clean	0.80	0.6497	0.5317	-0.1180	[-0.1506, -0.0901]	0.879
clean	0.90	0.6497	0.5225	-0.1271	[-0.1757, -0.0845]	1.000
gaussian_noise:all	0.40	0.6502	0.6502	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.50	0.6502	0.6502	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.60	0.6502	0.5478	-0.1025	[-0.1380, -0.0698]	0.800
gaussian_noise:all	0.66	0.6502	0.5267	-0.1235	[-0.1882, -0.0576]	1.000
gaussian_noise:all	0.70	0.6502	0.5267	-0.1235	[-0.1882, -0.0576]	1.000
gaussian_noise:all	0.80	0.6502	0.5267	-0.1235	[-0.1882, -0.0576]	1.000
gaussian_noise:all	0.90	0.6502	0.5267	-0.1235	[-0.1882, -0.0576]	1.000
max_attack:all	0.40	0.6330	0.6330	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.50	0.6330	0.6330	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.60	0.6330	0.6330	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.66	0.6330	0.5164	-0.1166	[-0.1831, -0.0394]	1.000
max_attack:all	0.70	0.6330	0.5164	-0.1166	[-0.1831, -0.0394]	1.000
max_attack:all	0.80	0.6330	0.5164	-0.1166	[-0.1831, -0.0394]	1.000
max_attack:all	0.90	0.6330	0.5164	-0.1166	[-0.1831, -0.0394]	1.000
zero_attack:all	0.40	0.6442	0.6442	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.50	0.6442	0.6442	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.60	0.6442	0.6442	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.66	0.6442	0.5199	-0.1243	[-0.1819, -0.0784]	1.000
zero_attack:all	0.70	0.6442	0.5199	-0.1243	[-0.1819, -0.0784]	1.000
zero_attack:all	0.80	0.6442	0.5199	-0.1243	[-0.1819, -0.0784]	1.000
zero_attack:all	0.90	0.6442	0.5199	-0.1243	[-0.1819, -0.0784]	1.000

asymmetry is sharp: the KS-drift signal that delivers the entire $+0.0506$ `zero_attack` gain on the label-aligned secondary benchmark misfires on legitimate inter-category variation in naturally paired data, places the mean reliability below τ on most batches, and the resulting adapted weighting consistently underperforms the static attention solution. The cleanest reading of the two results together is that a validation-derived drift detector cannot tell adversarial coherent score collapse apart from natural cross-category heterogeneity on this scale of paired data; the mechanism therefore needs a benchmark-conditioned drift threshold or a different drift signal before it can be deployed on natural paired evidence streams.

VII. Calibration and Explanation

A. Calibration

Table XXI reports calibration for the final configured seed. Static attention is slightly better than RGA on ECE and Brier score in that seed. Both attention variants remain much better calibrated than confidence-weighted mean fusion in the five-seed clean table. The reliability diagram is shown in Fig. 12.

B. Counterfactual Domain Attribution

Counterfactual domain attribution was computed for 400 test samples across configured seeds. Mean absolute probability changes were 0.0138 for fraud, 0.0172 for cyber, 0.0059 for behavior, and 0.0353 for text. Text has the largest average counterfactual impact, consistent with the strong text scorer in Table V; the per-domain magnitudes are visualized in Fig. 13.

VIII. Discussion and Limitations

A. What the Results Support

Across the two benchmarks the results support three scoped claims. First, on the label-aligned secondary benchmark ELARA-Bench-LA, the reliability gate improves all-domain coherent score-collapse robustness ($+0.0506$ `zero_attack`, $+0.0319$ `max_attack`) while preserving the saturated clean ranking. Second, the same mechanism is dominated by random

TABLE XIII
MVTEC 3D-AD RELIABILITY-COMPONENT ABLATION ON ALL-DOMAIN SCORE ATTACKS.

Variant	Attack	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
full	gaussian_noise	0.6495	0.5188	-0.1307	[-0.1870, -0.0825]	1.000
full	max_attack	0.6330	0.5164	-0.1166	[-0.1831, -0.0394]	1.000
full	zero_attack	0.6442	0.5199	-0.1243	[-0.1819, -0.0784]	1.000
no_ece	gaussian_noise	0.6495	0.5261	-0.1234	[-0.1877, -0.0809]	1.000
no_ece	max_attack	0.6330	0.5315	-0.1015	[-0.1612, -0.0402]	1.000
no_ece	zero_attack	0.6442	0.5299	-0.1143	[-0.1612, -0.0704]	1.000
no_gate	gaussian_noise	0.6495	0.5188	-0.1307	[-0.1870, -0.0825]	1.000
no_gate	max_attack	0.6330	0.5164	-0.1166	[-0.1831, -0.0394]	1.000
no_gate	zero_attack	0.6442	0.5199	-0.1243	[-0.1819, -0.0784]	1.000
no_ks	gaussian_noise	0.6495	0.6495	0.0000	[0.0000, 0.0000]	0.000
no_ks	max_attack	0.6330	0.6330	0.0000	[0.0000, 0.0000]	0.000
no_ks	zero_attack	0.6442	0.6442	0.0000	[0.0000, 0.0000]	0.000
no_sharpness	gaussian_noise	0.6495	0.5314	-0.1181	[-0.1633, -0.0796]	0.921
no_sharpness	max_attack	0.6330	0.5180	-0.1150	[-0.1803, -0.0383]	1.000
no_sharpness	zero_attack	0.6442	0.5188	-0.1253	[-0.1822, -0.0775]	1.000

TABLE XIV
MVTEC 3D-AD CALIBRATION AND COUNTERFACTUAL DOMAIN-ATTRIBUTION SUMMARY.

Metric	Static	RGA
ECE	0.0354	0.0402
Brier	0.1648	0.1698
CDA samples	–	400
CDA/ECE Spearman	–	–
CDA/ECE Spearman status	–	n/a (fewer than 3 domains)

TABLE XV
MVTEC 3D-AD REPRESENTATIVE FAILURE AND CONTRAST CASES.

Case	Label	Static p	RGA p	Sample
biggest_elara_win	0	0.5951	0.0520	mvtec3d_foam_train_good_150
biggest_elara_loss	1	0.7030	0.0329	mvtec3d_cable_gland_test_cut_0
elara_correct_static_wrong	0	0.5601	0.0505	mvtec3d_foam_train_good_151
static_correct_elara_wrong	1	0.6612	0.0319	mvtec3d_cable_gland_test_hole_
high_confidence_elara_failure	1	0.5319	0.0338	mvtec3d_cable_gland_test_threa

forest fusion on naturally paired MVTec 3D-AD across every condition including clean (random forest 0.959 vs static attention 0.650 vs RGA 0.610). Third, the contrastive mechanism-isolation results together identify a specific failure mode of validation-derived drift detectors: a KS-drift signal calibrated on validation can deliver gains on coherent adversarial collapse and simultaneously misfire on legitimate inter-category heterogeneity in naturally paired data. These are useful systems results, but they are bounded by the benchmark design and the current lightweight image-statistic scorer used on MVTec 3D-AD.

B. What the Results Do Not Support

The results do not support a broad claim that RGA improves multimodal fusion in general. On the primary naturally paired benchmark, RGA strictly hurts clean and adversarial ROC-AUC; on the secondary label-aligned benchmark, the gain is real but is bounded to coherent all-domain attacks and conditioned on a benchmark in which clean accuracy is near saturated. The results also do not establish that any of the included attention models matches random forest fusion on naturally paired MVTec 3D-AD; the score-level gap is large

TABLE XVI
MISSING-DOMAIN ABLATION. DELTA IS RGA MINUS STATIC ATTENTION.

Dropout	Static ROC	RGA ROC	Delta	Delta 95% CI
0.0	0.9999	0.9999	0.0000	[0.0000, 0.0000]
0.1	0.9995	0.9995	0.0000	[0.0000, 0.0000]
0.2	0.9981	0.9981	-0.0000	[-0.0000, 0.0000]
0.3	0.9964	0.9964	-0.0001	[-0.0002, 0.0000]
0.5	0.9784	0.9783	-0.0001	[-0.0004, 0.0000]

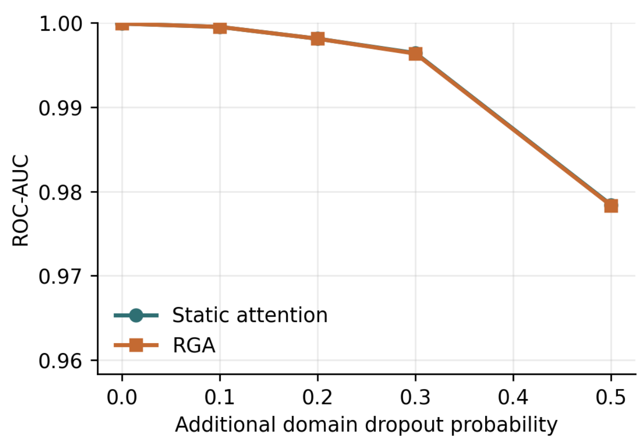

Fig. 9. ROC-AUC under additional missing-domain dropout.

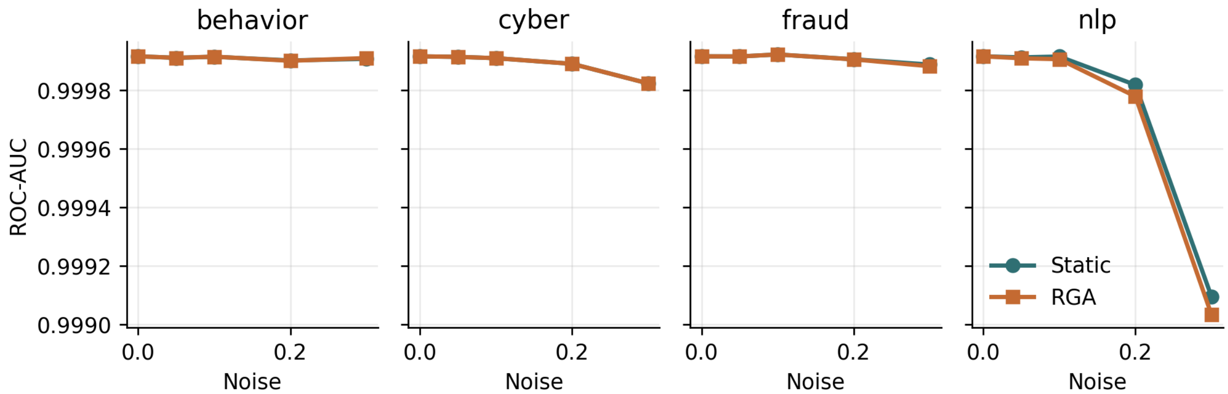

Fig. 10. Per-domain score-drift curves on the real-domain benchmark.

TABLE XVII
DRIFT ROBUSTNESS SUMMARY. HIGHER AREA IS BETTER.

Domain	Static	RGA	Delta	Seeds
behavior	0.3000 [0.3000, 0.3000]	0.3000 [0.3000, 0.3000]	0.0000	5
cyber	0.3000 [0.3000, 0.3000]	0.3000 [0.3000, 0.3000]	0.0000	5
fraud	0.3000 [0.3000, 0.3000]	0.3000 [0.3000, 0.3000]	-0.0000	5
nlp	0.2999 [0.2999, 0.2999]	0.2999 [0.2999, 0.2999]	-0.0000	5

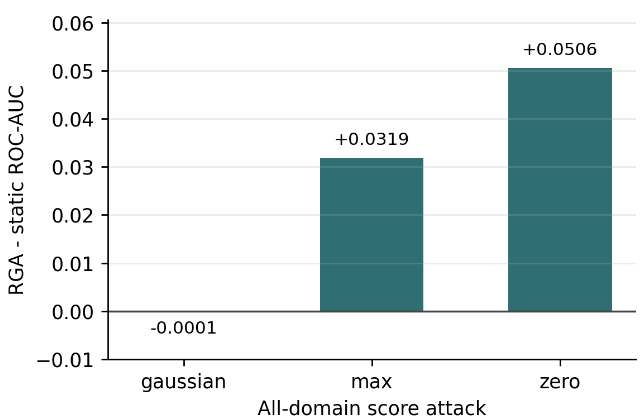

Fig. 11. All-domain adversarial ROC-AUC delta between RGA and static attention. Complete single-domain scenarios are reported in Table XVIII.

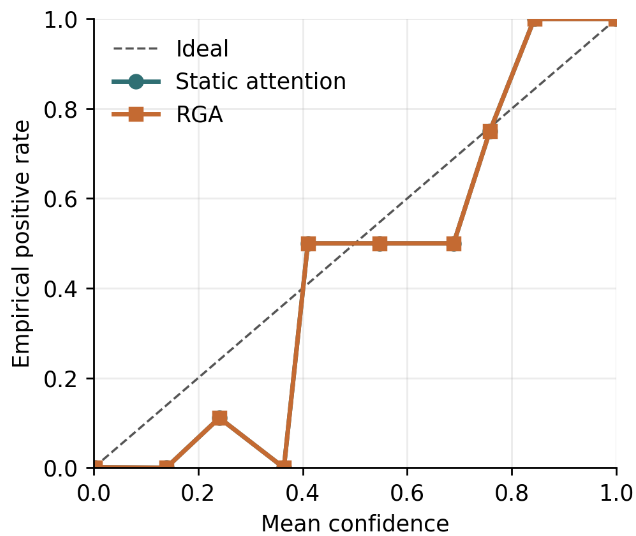

Fig. 12. Reliability diagram for static attention and RGA.

and persists across seeds.

C. Failure Analysis

The result artifact exports representative cases for the largest RGA win, largest RGA loss, cases where RGA fixes static attention, cases where static attention fixes RGA, and high-confidence RGA failures. Each case includes the sample identifier, label, static and RGA probabilities, per-domain scores, missing-domain mask, and reliability weights.

Representative cases are listed in Table XXII.

Across both benchmarks, the experiments expose four failure modes:

- the label-aligned construction makes the secondary clean split highly separable and obscures method differences,
- on naturally paired MVTec 3D-AD the reliability gate misfires on legitimate cross-category variation and degrades both clean and adversarial ROC-AUC,
- random forest fusion remains the strongest baseline on the paired benchmark by a clear margin,
- the lightweight image-statistic domain scorer caps the achievable ceiling on MVTec 3D-AD and is the natural next replacement target.

D. Internal Validity

The benchmark scripts are deterministic for configured seeds, and clean results are reported across five seeds. The runner repeats robustness ablations across configured seeds and stores confidence intervals. Residual risk remains if the

TABLE XVIII
ADVERSARIAL PERTURBATION BENCHMARK. DELTA IS RGA MINUS STATIC ATTENTION.

Attack	Target	Static ROC	RGA ROC	Delta	Delta 95% CI
gaussian_noise	all	0.9999	0.9998	-0.0001	[-0.0001, 0.0000]
gaussian_noise	behavior	0.9999	0.9999	-0.0000	[-0.0000, 0.0000]
gaussian_noise	cyber	0.9999	0.9999	0.0000	[0.0000, 0.0000]
gaussian_noise	fraud	0.9999	0.9999	0.0000	[0.0000, 0.0000]
gaussian_noise	nlp	0.9999	0.9999	-0.0000	[-0.0000, 0.0000]
max_attack	all	0.7401	0.7720	0.0319	[0.0050, 0.0617]
max_attack	behavior	0.9997	0.9997	-0.0000	[-0.0000, -0.0000]
max_attack	cyber	0.9972	0.9972	0.0000	[0.0000, 0.0000]
max_attack	fraud	0.9996	0.9996	0.0000	[0.0000, 0.0000]
max_attack	nlp	0.9840	0.9843	0.0003	[-0.0009, 0.0016]
zero_attack	all	0.7212	0.7718	0.0506	[0.0315, 0.0681]
zero_attack	behavior	0.9997	0.9997	0.0000	[-0.0000, 0.0000]
zero_attack	cyber	0.9980	0.9980	0.0000	[0.0000, 0.0000]
zero_attack	fraud	0.9995	0.9995	0.0000	[0.0000, 0.0000]
zero_attack	nlp	0.9713	0.9716	0.0003	[-0.0004, 0.0009]

Fig. 13. Mean absolute counterfactual domain-attribution impact.

stress distributions are too synthetic relative to deployment failures.

E. Construct Validity

ELARA-Bench-LA measures real-domain score fusion, not naturally paired multimodal incident detection. The MVTec 3D-AD path addresses natural RGB/3D pairing, but the current local run is a small bagel-subset smoke benchmark with lightweight features. This remains the largest construct-validity threat. The paper therefore avoids claiming that the model has learned causal or temporal relationships among fraud, cyber, behavior, and text domains, or that the paired path is competitive with specialized RGB–3D anomaly detectors.

F. External Validity

The datasets are useful but not sufficient for deployment claims. A production system would need domain-specific label definitions, temporal splits, deployment-time calibration monitoring, privacy review, and dataset license review.

G. Statistical Validity

Five clean seeds provide only limited statistical support, especially on a near-saturated split. This draft therefore reports seed-level confidence intervals for clean PR-AUC and F1 and treats stress-test deltas as the primary diagnostic. DeLong tests are still used where assumptions hold [14], but the paper avoids treating a single significance test as proof of broad superiority.

IX. REPRODUCIBILITY

The reproducibility flow has three scripted steps:

1) build the real-domain fusion input table,
2) run the real fusion benchmark with the YAML configuration,
3) generate tables, figures, and the PDF manuscript.

The implementation uses the local scripts under `src/scripts/`, the configurations `configs/attention_real_fusion.yaml` and `configs/attention_mvtec3d_fusion.yaml`, result artifacts under `experiments/fusion/`, and generated paper assets under `docs/research/`. The paired benchmark entry point is `src/scripts/prepare_`

TABLE XIX
RELIABILITY-GATE THRESHOLD SWEEP. DELTA IS RGA MINUS STATIC ATTENTION; ADAPTATION RATE IS THE FRACTION OF EVALUATED SAMPLES USING THE RELIABILITY-AWARE PATH.

Condition	τ	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
clean	0.40	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
clean	0.50	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
clean	0.60	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
clean	0.66	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
clean	0.70	0.9999	0.9999	-0.0000	[-0.0000, 0.0000]	0.320
clean	0.80	0.9999	0.9999	-0.0000	[-0.0001, 0.0000]	0.960
clean	0.90	0.9999	0.9999	-0.0000	[-0.0001, 0.0000]	1.000
gaussian_noise:all	0.40	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.50	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
gaussian_noise:all	0.60	0.9999	0.9998	-0.0000	[-0.0001, -0.0000]	0.680
gaussian_noise:all	0.66	0.9999	0.9998	-0.0001	[-0.0002, -0.0000]	1.000
gaussian_noise:all	0.70	0.9999	0.9998	-0.0001	[-0.0002, -0.0000]	1.000
gaussian_noise:all	0.80	0.9999	0.9998	-0.0001	[-0.0002, -0.0000]	1.000
gaussian_noise:all	0.90	0.9999	0.9998	-0.0001	[-0.0002, -0.0000]	1.000
max_attack:all	0.40	0.7401	0.7401	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.50	0.7401	0.7401	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.60	0.7401	0.7401	0.0000	[0.0000, 0.0000]	0.000
max_attack:all	0.66	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
max_attack:all	0.70	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
max_attack:all	0.80	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
max_attack:all	0.90	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
zero_attack:all	0.40	0.7212	0.7212	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.50	0.7212	0.7212	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.60	0.7212	0.7212	0.0000	[0.0000, 0.0000]	0.000
zero_attack:all	0.66	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
zero_attack:all	0.70	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
zero_attack:all	0.80	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
zero_attack:all	0.90	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000

`mvtec3d_fusion_benchmark.py`. Targeted regression tests cover attention masking, statistics, baselines, reliability estimation, counterfactual explanation, paired-benchmark metadata, and result aggregation.

X. RELATIONSHIP TO COMPANION THESIS CHAPTER

A longer, thesis-style treatment of this work is maintained alongside the manuscript as `docs/research/THESIS_CHAPTER_v1.tex`, compiled to `output/pdf/THESIS_CHAPTER_v1.pdf`. The conference manuscript and the thesis chapter share the same code base, configurations, and result artifacts; what differs is the audience and the level of reflection. The thesis chapter is longer, contains a full Background section, expanded construct-validity and statistical-validity discussion, an explicit "what did not work" subsection, and explicit framing of failure cases as evidence rather than appendix material. The conference manuscript prioritizes the contrastive headline result – gate-helps-on-label-aligned versus gate-hurts-on-natural-paired – and uses the thesis chapter as the place to develop those threads in more detail.

The two documents are kept consistent through a single asset pipeline. The benchmark JSONs (`experiments/fusion/craf_real_results.json` and `experiments/fusion/mvtec3d_results.json`) generate the same `rga_*.tex` and `mvtec3d_*.tex` tables and `*.png` figures, which both manuscripts `input` or `includegraphics`. Changes to the experimental evidence therefore propagate to both documents without divergence. Readers who want the systems-paper framing should stay with this manuscript; readers who want a thesis-style discussion (including the extended background, threats-to-validity, and forward-looking research agenda) should read the companion chapter.

XI. CONCLUSION

ELARA presents a reliability-aware anomaly-fusion pipeline with domain experts, unified score schema,

TABLE XX
RELIABILITY-COMPONENT ABLATION ON ALL-DOMAIN SCORE ATTACKS. DELTA IS RGA MINUS STATIC ATTENTION.

Variant	Attack	Static ROC	RGA ROC	Delta	Delta 95% CI	Adapt rate
full	gaussian_noise	0.9999	0.9997	-0.0002	[-0.0005, -0.0000]	1.000
full	max_attack	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
full	zero_attack	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
no_ece	gaussian_noise	0.9999	0.9996	-0.0003	[-0.0005, -0.0001]	1.000
no_ece	max_attack	0.7401	0.7858	0.0457	[0.0129, 0.1067]	1.000
no_ece	zero_attack	0.7212	0.7841	0.0629	[0.0493, 0.0808]	1.000
no_gate	gaussian_noise	0.9999	0.9997	-0.0002	[-0.0005, -0.0000]	1.000
no_gate	max_attack	0.7401	0.7720	0.0319	[0.0050, 0.0617]	1.000
no_gate	zero_attack	0.7212	0.7718	0.0506	[0.0315, 0.0681]	1.000
no_ks	gaussian_noise	0.9999	0.9999	0.0000	[0.0000, 0.0000]	0.000
no_ks	max_attack	0.7401	0.7401	0.0000	[0.0000, 0.0000]	0.000
no_ks	zero_attack	0.7212	0.7212	0.0000	[0.0000, 0.0000]	0.000
no_sharpness	gaussian_noise	0.9999	0.9997	-0.0002	[-0.0005, -0.0000]	1.000
no_sharpness	max_attack	0.7401	0.7808	0.0408	[0.0123, 0.0811]	1.000
no_sharpness	zero_attack	0.7212	0.7759	0.0547	[0.0382, 0.0716]	1.000

TABLE XXI
CALIBRATION AND COUNTERFACTUAL DOMAIN-ATTRIBUTION SUMMARY.

Metric	Static	RGA
ECE	0.0038	0.0038
Brier	0.0032	0.0032
CDA samples	–	400
CDA/ECE Spearman	–	0.052
CDA/ECE Spearman status	–	computed

TABLE XXII
REPRESENTATIVE FAILURE AND CONTRAST CASES FROM THE LATEST CONFIGURED SEED.

Case	Label	Static p	RGA p	Sample
biggest_elara_win	0	0.0001	0.0001	real_fusion_test_000000
biggest_elara_loss	1	0.9998	0.9998	real_fusion_test_001072
high_confidence_elara_failure	1	0.0470	0.0470	real_fusion_test_001517

masked attention, reliability-gated adaptation, counterfactual explanation, benchmark construction, baselines, ablations, calibration, and failure analysis. The strongest measured claim is that RGA preserves clean attention performance, improves over confidence-weighted mean fusion, and helps under severe all-domain score collapse. The main remaining research gap is full naturally paired multimodal validation with stronger RGB–3D domain experts.

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